INCH-POUND

MIL-PRF-81757/1E
11 September 2013
SUPERSEDING
MIL-PRF-81757/1D
1 March 1999

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, CELL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-81757.

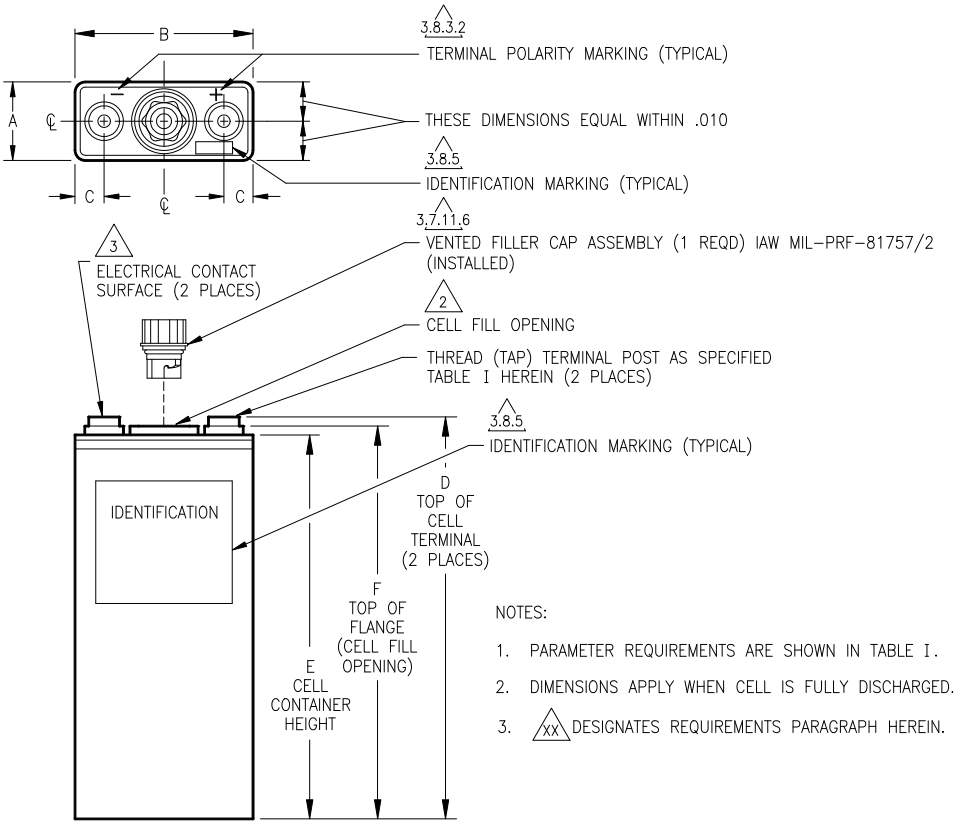
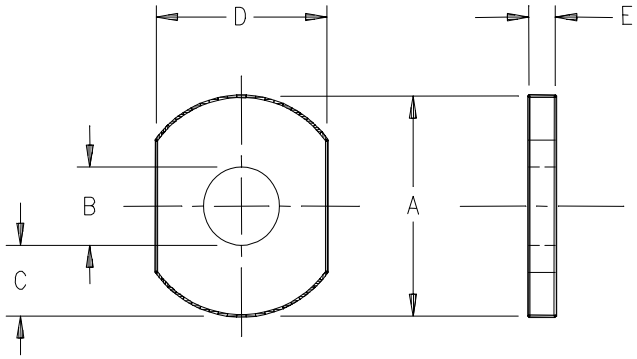


FIGURE 1. Dimensions and configuration for cells (PINs M81757/1-2, M81757/1-4, and M81757/1-6).

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TABLE I. Parameter requirements for cells (PINs M81757/1-2, M81757/1-4, and M81757/1-6).

PART OR IDENTIFYING NUMBER (PIN)	DIMENSIONS IN INCHES (SEE FIGURE 1)						TERMINAL POST (SEE FIGURE 1)		CAPACITY AMP HRS (MIN)		WEIGHT POUNDS (MAX)
	A +.015 -.025	B +.015 -.025	C ±.010	D ±.020	E Ref	F Max	THREAD INTERNAL NO	THREAD DEPTH INCH (MIN)	OTHER THAN INITIAL (RATED)	INITIAL	
M81757/1-2	1.065	2.414	.395	6.525	6.280	6.388	8-32 UNC-3B	.466	10.0	12.5	1.30
M81757/1-4	1.070	3.175	.650	7.869	7.606	7.716	10-32 NF-3B	.500	20.0	25.0	2.30
M81757/1-6	1.375	3.130	.540	8.932	8.678	8.838	10-32 NF-3B	.500	30.0	37.5	3.40



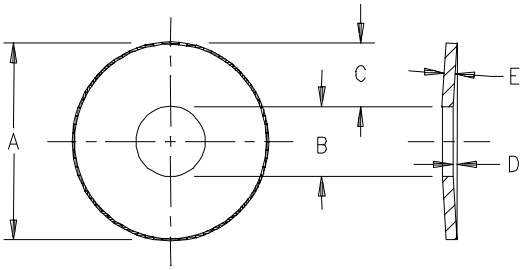
NOTES:
1. PARAMETER DIMENSIONS ARE SHOWN IN TABLE II.

FIGURE 2. Washers (PINs M81757/1-7A and M81757/1-7B).

TABLE II. Dimensions for washers (PINs M81757/1-7A and M81757/1-7B).

PART OR IDENTIFYING NUMBER (PIN)	DIMENSIONS IN INCHES (SEE FIGURE 2)				
	A ±.030	B ±.010	C ±.020	D ±.030	E ±.010
M81757/1-7A (Washer)	.500	.178	.161	.400	.062
M81757/1-7B (Washer)	.750	.203	.273	.600	.062

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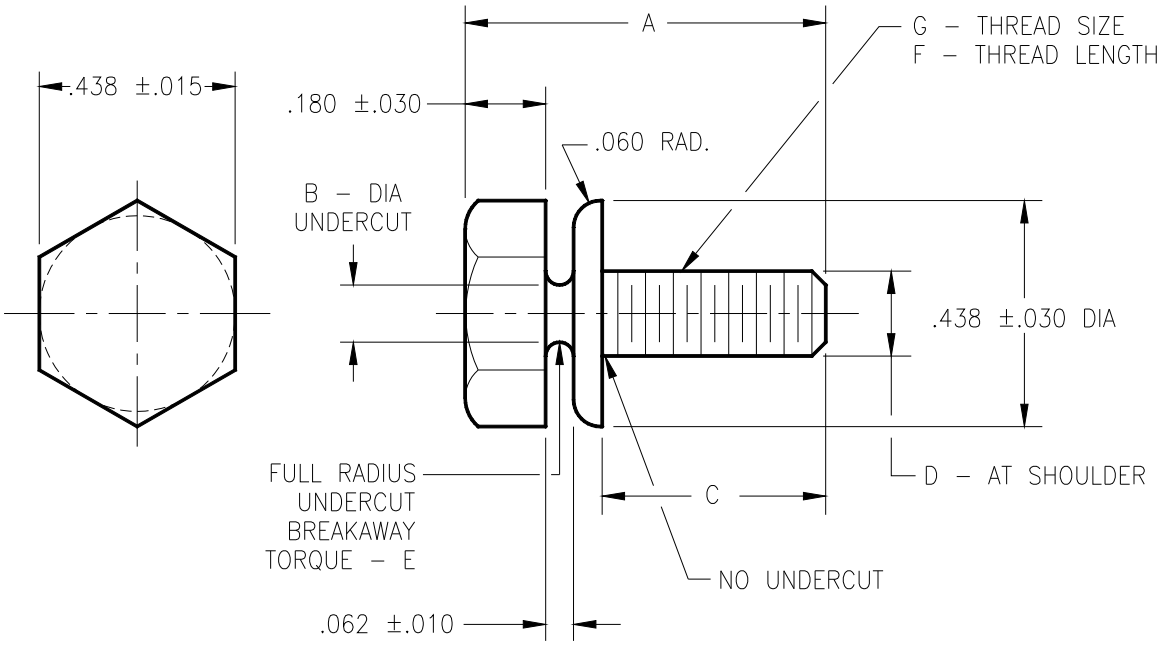
NOTES:
1. PARAMETER DIMENSIONS ARE SHOWN IN TABLE III.

FIGURE 3. Belleville springs (PINs M81757/1-8A and M81757/1-8B).

TABLE III. Dimensions for Belleville springs (PINs M81757/1-8A and M81757/1-8B).

PART OR IDENTIFYING NUMBER (PIN)	DIMENSIONS IN INCHES (SEE FIGURE 3)				
	A ±.030	B ±.010	C ±.020	D ±.005	E ±.005
M81757/1-8A (Belleville spring)	.500	.178	.161	.011	.028
M81757/1-8B (Belleville spring)	.561	.203	.179	.025	.028

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NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. PARAMETER REQUIREMENTS ARE SHOWN IN TABLE IV.

FIGURE 4. Terminal screws (breakaway head) (PINs M81757/1-9 and M81757/1-10).

TABLE IV. Parameter requirements for terminal screws (PINs M81757/1-9 and M81757/1-10).

PART OR IDENTIFYING NUMBER (PIN)	A	B	C	D	E	F	G
	REF (INCH)	±.002 (INCH)	MAX (INCH)	±.004 (INCH)	HEAD BREAK TORQUE (IN LBS)	THREAD FULL LENGTH (INCH)	THREAD EXTERNAL NO
M81757/1-9	.766	.112	.466	.164	38 Min 48 Max	.400 Min	8-32 UNC-3A
M81757/1-10	.806	.128	.500	.188	55 Min 72 Max	.440 Min	10-32 NF-3A

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REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the items covered by this specification sheet shall be as shown below. Each PIN within a category differs in terms of dimensions and other parameters from the other PINs in that category. Each item shall conform to the requirements of the indicated figures and tables.

a. Cells. PINs M81757/1-2, M81757/1-4, and M81757/1-6 on figure 1 and in table I.

b. Washers. PINs M81757/1-7A and M81757/1-7B on figure 2 and in table II.

c. Belleville springs. PINs M81757/1-8A and M81757/1-8B on figure 3 and in table III.

d. Terminal screws. PINs M81757/1-9 and M81757/1-10 on figure 4 and in table IV.

2. The cell fill opening shall accommodate and interface with all installation features of the vented filler cap supplied in accordance with 3.7.11.6 herein. This shall include:

a. The mechanisms for attaching (tightening and locking) and removing the cap.

b. The methods used for creating a seal (air and liquid) at the fill opening (see 3.7.11.7 and 3.10.13 of the general specification).

3. The roughness of the surface finish of the electrical contact area of each terminal post shall be not greater than 63 microinches high and not greater than 0.02 inch wide. Waviness shall be not greater than 0.005 inch high.

4. MIL-PRF-81757 variance. The items shall comply with MIL-PRF-81757 except as follows.

4.1 Modify the following paragraph:

3.7.11.6 Vented filler cap. Insert after the first sentence: "The particular MIL-PRF-81757/2 vented filler cap (such as PIN M81757/2-1, M81757/2-2) to be installed on the cell shall be as follows:

"a. Unless otherwise indicated, when a cell is purchased separately from a battery, install the M81757/2-1 vented filler cap (see 6.2).

"b. When a cell is supplied as a component installed in a MIL-PRF-81757 battery, install the vented filler cap that is required by the battery's specification sheet."

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4.2 Add the following paragraph:

"3.7.11.9 Cell fasteners. Terminal screws, washers, and Belleville springs are not required for cells that are procured separately from complete batteries. These items shall meet the requirements of 3.10.2 and 3.10.19 (see 6.10.1). In addition, the tensile strength of the terminal screws (PINs M81757/1-9 and M81757/1-10) shall be 130,000 ± 20,000 pounds per square inch (psi) as determined by subjecting the items to 4.5.31."

4.3 Modify the following paragraphs:

3.8.3.2 Cells. Insert as a new first sentence: "The cell's terminal polarity markings shall be located as shown on figure 1 of the specification sheet."

3.8.5 Cell identification. Insert after the third sentence: "The cell's identification markings shall be located as shown on figure 1 of the specification sheet."

4.4 Add the following paragraph:

"4.5.31 Head breakaway torque test for M81757/1-9 and M81757/1-10 terminal screws. Secure the terminal screw's body in a fixed position and subject the terminal screw's head to a torque not greater than 20 inch-pounds. Gradually increase the applied torque until the terminal screw's head separates from the terminal screw's body. The applied torque (inch-pounds) at separation (breakaway) shall comply with the head breakaway torque requirement specified in 3.7.11.9 and table IV of the specification sheet."

4.5 Modify the following paragraphs and table:

6.2.d Acquisition requirements. Insert after the first sentence: "Cells must be ordered in multiples of 19 cells. Cells must be packaged in groups of exactly 19 cells. Each cell within a 19 cell group must be part of the same inspection lot and bear the same date/lot code (see 3.8.2). Specify the MIL-PRF-81757/2 vented filler cap to be used (see 3.7.11.6 as modified herein)."

6.10.1 Corrosive nature of potassium hydroxide. Add as a new last sentence: "Components manufactured from stainless steel 303 have passed the qualification tests in the past."

6.15 PIN supersession data. Delete text and substitute: "Cell PINs M81757/1-2, M81757/1-4, and M81757/1-6 supersede the cells shown in tables IX and X."

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TABLE IV. Qualification inspection of components and materials. For "Metal parts," add the salt fog test (3.10.19 and 4.5.19) and the head breakaway torque test for M81757/1-9 and M81757/1-10 terminal screws (3.7.11.9 and 4.5.31).

TABLE IX. Battery and cell supersession data. Delete the title and contents and substitute the following table:

TABLE IX. M81757/1-2, M81757/1-4, and M81757/1-6 supersession data (see notes).

Superseding PIN (cell)	Superseded PINs (cells)
M81757/1-2	BB-599 MS27307-1 MS27307-4 MS90321-66 MS90321-72 M81757/1-1
M81757/1-4	MS27307-2 MS27307-5 MS90321-68 MS90321-74 MS90321-77 MS90321-78 M81757/1-3
M81757/1-6	MS27307-3 MS27307-6 MS90321-69 MS90321-75 M81757/1-5

NOTES:

1. When purchasing for use by military departments that use only limited-repair nickel-cadmium batteries, a MIL-PRF-81757 battery shall be substituted in accordance with the following for each quantity of 19 or fewer cells.

M81757/1-2 cell	-	Substitute battery M81757/7-3 or
M81757/1-4 cell	-	Substitute battery M81757/8-5, M81757/11-3, whichever is the correct design for the application.
M81757/1-6 cell	-	Substitute battery M81757/9-3 or M81757/13-1, whichever is the correct design for the application.

2. All variations of each MS90321 cell are included; for example, -66, -66W, and -66R.

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Replace TABLE VIII. Group C inspection with the following:

Test	Requirement paragraphs	Test paragraph	Sample
1. Inspection for Damage	3.10.1	4.5.1	All
2. Visual and Mechanical	3.10.6 3.6 thru 3.8 Table II	4.5.6	All
3. Dimensions and Weight (weight only)	3.10.8 3.10.9	4.5.8	All
4. Initial Capacity Discharge	3.10.14	4.5.13	All
5. Cycling Test	3.10.23	4.5.23	1
6. Humidity and Charge Retention	3.9 3.10.15	4.5.14	2
7. Altitude	3.9 3.10.18	4.5.18	3
8. Temperature Rise and Float	3.9 3.10.9 3.10.24	4.5.24.2	3
9. Medium-Rate, Operating Position	3.9 3.10.21	4.5.21	2
10. Internal Air Pressure Test	3.9 3.10.26	4.5.26	All

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4.6 Add the following table:

TABLE X. Superseded (replaced) cells.

Superseded PIN (cell)	General or procurement specification of superseded PIN	Superseding PIN (cell)	Superseded PIN (cell)	General or procurement specification of superseded PIN	Superseding PIN (cell)
BB-599/A	---	M81757/1-2	M81757/1-5	MIL-PRF-81757	M81757/1-6
BB-601	---	M81757/1-4	MS90321-66	MIL-B-8565	M81757/1-2
B-600/A	---	M81757/1-6	MS90321-68	MIL-B-8565	M81757/1-4
MS27307-1	MIL-B-26220	M81757/1-2	MS90321-69	MIL-B-8565	M81757/1-6
MS27307-2	MIL-B-26220	M81757/1-4	MS90321-72	MIL-B-8565	M81757/1-2
MS27307-3	MIL-B-26220	M81757/1-6	MS90321-74	MIL-B-8565	M81757/1-4
MS27307-4	MIL-B-26220	M81757/1-2	MS90321-75	MIL-B-8565	M81757/1-6
MS27307-5	MIL-B-26220	M81757/1-4	MS90321-77	MIL-B-8565	M81757/1-4
MS27307-6	MIL-B-26220	M81757/1-6	MS90321-78	MIL-B-8565	M81757/1-4
M81757/1-1	MIL-PRF-81757	M81757/1-2	MS90321-79	MIL-B-8565	M81757/1-4
M81757/1-3	MIL-PRF-81757	M81757/1-4			

NOTE: All variations of each MS90321 cell are included; for example -66, -66W, and -66R.

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Referenced documents: In addition to MIL-PRF-81757, this document references the following:

MIL-PRF-81757/2

5. Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - AS
Air Force - 11

Review activities:
Army - AV
Air Force - 71, 99
DLA - CC, GS

Preparing activity:
Navy - AS

Agent:
Navy - SH
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